

Built from: **Aman's AI Journal Interview Questions PDF**.
Goal: high-density review in **10-15 minutes** before interviews.

0) 2-minute review strategy

1. Read sections **1, 2, 5** first (highest interview hit rate).
2. Then skim **3 and 4** for depth probes.
3. Use section **8** for formula recall and section **9** for rapid-fire.
4. If you get stuck in interview: answer with **definition -> intuition -> equation -> trade-off -> failure mode**.

1) Core ML + DL fundamentals (must-answer cleanly)

Bias-variance tradeoff

- ? **Bias**: underfit, high training/test error.
- ? **Variance**: overfit, low train error, high test error.
- ? Control with model capacity, regularization, data size, augmentation.
- ? Staff-level answer includes **which knob** you would turn first and why.

L1 vs L2 regularization

- ? L1 penalty: $\lambda \sum_i |w_i|$ -> sparse weights, feature selection.
- ? L2 penalty: $\lambda \sum_i w_i^2$ -> smooth shrinkage, stability.
- ? Use L2 as default for dense signals; L1 when sparsity/selection matters.

Why random initialization?

- ? Same initialization for all neurons in a layer -> identical gradients -> neurons remain symmetric -> no diversity learned.
- ? Random init breaks symmetry and allows specialized feature learning.

Why MSE is poor for classification

- ? MSE with sigmoid can yield slow gradients in saturated regions.
- ? Cross-entropy gives stronger, better-shaped gradients for probabilistic classification.

Batch / stochastic / mini-batch GD

- ? Batch GD: stable but slow per update.
- ? SGD: noisy but can escape sharp minima/saddles.
- ? Mini-batch: practical default; best throughput and convergence balance.

2) Transformers / LLMs / GenAI (very common senior probes)

Self-attention in one line

- ? Computes context-aware token representations by weighted aggregation of all token values.

Why Q, K, V?

- ? $(QK^{\top})$ computes relevance scores (who should attend to whom).
- ? (V) holds payload to aggregate.
- ? Separating (K) and (V) decouples **matching** from **content**.

Attention formula

- ? $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\text{top}}}{\sqrt{d_k}}\right)V$
- ? $\sqrt{d_k}$ scaling prevents large dot products from saturating softmax.

Transformer drawbacks

- ? $O(n^2)$ attention cost (memory/latency bottleneck on long context).
- ? Data-hungry; can hallucinate; inference expensive.
- ? Mitigations: sparse/linear attention, retrieval, chunking, KV-cache optimizations.

Temperature / top-k / top-p

- ? Temperature controls sharpness of token distribution.
- ? top-k truncates to k most likely tokens.
- ? top-p (nucleus) truncates to smallest set with cumulative prob $\geq p$.
- ? Higher values $\Rightarrow$ more diversity, less determinism.

Why same prompt can produce different outputs

- ? Sampling randomness + temperature/top-k/top-p + seed differences + backend nondeterminism.
- ? At near-zero temperature and fixed seed, outputs are more reproducible.

DDPM vs DDIM

- ? DDPM: stochastic reverse process; more diverse, slower sampling.
- ? DDIM: deterministic/implicit trajectory; faster sampling, often less diversity.

GAN mode collapse

- ? Generator maps many latent points to limited outputs (low diversity).
- ? Mitigate with improved objectives (WGAN-GP), minibatch discrimination, better D/G balance.

RLHF at high level

- ? SFT $\rightarrow$ preference data $\rightarrow$ reward model $\rightarrow$ PPO/DPO-style optimization.
- ? Key risk: reward hacking / alignment tax / preference noise.

3) Training pathologies + debugging checklist

Overfitting mitigation stack (in order)

1. Data quality + leakage check.
2. Train/val split sanity (no overlap, proper time split if needed).
3. Regularization (weight decay, dropout, label smoothing).
4. Early stopping + learning-rate schedule.
5. Augmentation / more data.
6. Reduce model capacity or freeze lower layers in fine-tuning.

Vanishing gradients

- ? Deep nets with saturating activations or poor initialization can lose gradient signal.
- ? Fixes: ReLU-family activations, residual connections, normalization, careful initialization.

Batch norm: why it helps

- ? Stabilizes optimization landscape and gradient flow.
- ? Enables larger learning rates and faster convergence.

Validation loss up while accuracy up

- ? Possible when confidence calibration worsens: more correct labels but overconfident wrong predictions inflate log loss.

If test performance underperforms

? Check: leakage, distribution shift, label noise, class imbalance, metric mismatch, insufficient hyperparameter search, data preprocessing mismatch.

4) Classical ML, stats, probability (still asked frequently)

Pearson vs Spearman

? Pearson: linear correlation on raw values.

? Spearman: rank correlation (monotonic relationships, robust to outliers/non-normality).

CLT use case

? Means/sums of many independent samples approximate normal; allows confidence intervals/tests even if raw data non-normal (with caveats).

Cross-validation

? Estimates generalization robustly on small/medium datasets.

? Use **stratified** for imbalance, **grouped/time-series CV** when IID assumption breaks.

Random forest vs gradient boosting

? RF: bagging, parallel trees, variance reduction, robust baseline.

? GBM/XGBoost: sequential residual fitting, lower bias, often better accuracy, more tuning-sensitive.

K-means convergence nuance

? Objective (within-cluster SSE) monotonically decreases; converges to a **local** optimum.

? Not guaranteed unique/global optimum; sensitive to initialization.

Imbalanced classes

? Use class weights, focal loss, threshold tuning, resampling, PR-AUC/F1/recall@precision targets.

? Avoid reporting accuracy alone.

Long-tail data

? Majority classes dominate gradients.

? Use reweighting, resampling, decoupled head training, focal/class-balanced losses.

5) Experimentation and decision quality (staff differentiator)

A/B testing pitfalls

? Sample ratio mismatch, peeking, multiple comparisons, novelty effects, interference between users.

? Staff answer includes pre-registered metrics + guardrails + stopping rule.

P-value in plain language

? Probability of observing data this extreme (or more) **assuming null is true**.

? Not the probability that null is true.

Metric selection

? Match metric to business objective and failure costs.

? Track primary metric + guardrails (latency, safety, fairness, cost).

Selection bias

? Data collection process distorts representativeness.

? Fix with better sampling design, randomization, stratification, and post-stratification checks.

6) Applied ML engineer/system view (what senior loops test)

On-device deployment

- ? Optimize with quantization, pruning, distillation, operator fusion, batching policy.
- ? Track latency p95/p99, memory footprint, thermal/power constraints.

Distribution shift

- ? Types: covariate shift, label shift, concept drift.
- ? Build drift monitors + retraining triggers + rollback plan.

Uncertainty for regression

- ? Aleatoric (data noise) vs epistemic (model uncertainty).
- ? Methods: quantile regression, ensembling/MC dropout, conformal prediction.

Hallucination mitigation

- ? Retrieval grounding, constrained decoding, tool verification, abstain option, self-consistency with verifier.

7) SQL/Python coding mini-cheats

SQL order of execution (logical)

'FROM -> JOIN -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY -> LIMIT'

Common SQL gotchas

- ? 'WHERE' filters rows pre-aggregation; 'HAVING' filters groups post-aggregation.
- ? 'BETWEEN' is inclusive; 'IN' matches set membership.
- ? Prefer explicit joins over comma joins.

Python quick reminders

- ? 'list': mutable; 'tuple': immutable/hashable (if elements hashable).
- ? For set operations/intersection at scale, convert to 'set' first.
- ? For duplicates: hash map / 'collections.Counter' / sort+scan.

8) Formula flashcards (memorize)

- ? **Cross-entropy (binary)**: $-(y \log p + (1-y) \log(1-p))$
- ? **L2-regularized loss**: $L = L_{\text{data}} + \lambda \sum_i w_i^2$
- ? **L1-regularized loss**: $L = L_{\text{data}} + \lambda \sum_i |w_i|$
- ? **Softmax**: $\sigma(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$
- ? **Bayes rule**: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- ? **Precision**: $TP / (TP + FP)$
- ? **Recall**: $TP / (TP + FN)$
- ? **F1**: $2PR / (P + R)$
- ? **Attention**: $\text{softmax}(QK^{\top} / \sqrt{d_k})V$
- ? **K-means objective**: $\sum_k \sum_{i \in C_k} |x_i - \mu_k|^2$

9) Rapid-fire (speak each in ≤ 20 seconds)

1. Why random initialization? -> symmetry breaking.
2. Why mini-batch over full batch? -> throughput + stable enough gradients.
3. Pearson vs Spearman? -> linear vs rank-monotonic.
4. Why CE over MSE for classification? -> better gradients/probabilistic fit.
5. What is overfitting? -> memorization; poor generalization.
6. How to detect leakage? -> implausibly high CV, feature availability audit, time-based holdout.
7. Random forest vs boosting? -> variance reduction vs bias reduction.
8. Why Q/K/V? -> relevance scoring separated from value aggregation.
9. Why transformer expensive? -> quadratic attention.
10. Why outputs vary for same prompt? -> sampling/randomness/backend nondeterminism.
11. What is mode collapse? -> generator loses diversity.
12. What is focal loss doing? -> downweights easy examples, focuses hard/minority samples.
13. K-means guaranteed global optimum? -> no, local optimum.
14. What is label smoothing? -> reduce overconfidence, improve calibration.
15. What is CLT and why useful? -> sampling mean approx normal for inference.
16. What is selection bias? -> non-representative sampling; invalid conclusions.
17. What is batch norm benefit? -> stabilizes and speeds training.
18. What is concept drift? -> $\mathbb{P}(y|x)$ changes over time.
19. How to choose eval metric under imbalance? -> PR-AUC/recall@precision + business-cost thresholds.
20. Staff-level escalation trigger? -> when metric gain conflicts with safety/cost/latency guardrails.

10) 30-second staff answer template (use for any hard question)

1. **Define** the concept precisely.
2. Give **one intuition/example**.
3. Add **one equation/metric**.
4. State **trade-offs** and failure modes.
5. End with **how you'd validate** in production.

This format avoids rambling and signals senior depth quickly.